FYP MEETING LOG

Supervisor Meeting 2 of 3

Agent-Based Game Analysis in Unity Based Upon Curiosity and RND Exploration Using PPO and SAC
Student: Raza Ullah Khan (NUM-BSCS23-28) | Supervisor: Dr. Malik M Ali Shahid | Co-Supervisor: Mr. Abdul Rafay | Industrial Partner: Inspired Square

Meeting No.	04 (Supervisor Meeting 2)
Meeting With	Supervisor - Dr. Malik M Ali Shahid
Date	Tuesday, 27 October 2026
Time	10:00 AM - 10:30 AM
Duration	30 minutes
Mode / Venue	Supervisor's office, Namal University Mianwali
Attendees	Raza Ullah Khan (Student); Dr. Malik M Ali Shahid (Supervisor)
Milestone	FN-02 (Month 1, Weeks 3-4): Hummingbirds 3D Flight Area & Flower Setup

1. Purpose of the Meeting

To go through the main papers, fix some references in the proposal, and define clearly how each result will be measured before any training starts.

2. Agenda

1. Short summary of the key papers.
2. Corrections needed in the proposal references.
3. Clear definition of every metric.
4. How to combine results from different seeds.

3. What We Discussed

- **PPO and SAC.** PPO (Schulman et al., 2017) updates the policy in small, safe steps and is usually stable. SAC (Haarnoja et al., 2018) reuses old experience and tries to keep its actions varied, so it can learn with fewer samples. Zhang et al. (2024) compared both in Unity ML-Agents: SAC learned faster but was less stable.
- **Curiosity and RND.** Curiosity (Pathak et al., 2017) gives a bonus when the agent cannot predict what happens next. RND (Burda et al., 2018) gives a bonus when a small network cannot copy a fixed random network on a new state. Both push the agent to visit new places.
- **Reference fixes.** I found that some citations in the proposal point to the wrong paper. SAC should cite Haarnoja et al. (2018), RND should cite Burda et al. (2018), and the Zhang et al. (2024) Unity paper is missing from the list. Sir asked me to fix these before the final submission of the document.
- **Main metric.** The main result is the extrinsic return - the normal game reward only, without the exploration bonus. This is important; otherwise Curiosity and RND would look better just because they get extra reward.
- **Other metrics.** Task progress (nectar collected per episode), how fast learning reaches a good level, stability across seeds, and exploration coverage if we can measure it in a simple way.

4. Small Implementation Part

We wrote down a clear definition for every metric, and I showed a small helper that will turn the per-seed results into one line of the results table.

Metric	Simple meaning	How we measure it

Extrinsic return (main)	How well the agent does the real task	Mean episode reward from the game only, last part of training
Task progress	How much nectar it collects	Nectar collected per episode
Sample efficiency	How quickly it learns	Training steps needed to reach 80% of its best average
Stability	How much results jump	Standard deviation across the 5 seeds
Coverage (extra)	How much of the area it visits	Share of 3D grid cells visited (only if practical)

```
import numpy as np

def summarize(final_returns):
    # one value per seed  ->  mean and standard deviation
    arr = np.array(final_returns, dtype=float)
    return arr.mean(), arr.std(ddof=1)

def steps_to_threshold(steps, rewards, frac=0.8):
    # first training step where the reward reaches 80% of its best value
    target = frac * max(rewards)
    for s, r in zip(steps, rewards):
        if r >= target:
            return s
    return None
```

5. Decisions Taken

- Extrinsic return (without the intrinsic bonus) is the main metric for all six conditions.
- Sample efficiency is measured as steps to reach 80% of the best average reward.
- Coverage is an extra diagnostic only; it will not decide the comparison.
- Proposal references will be corrected (SAC, RND, and the missing Zhang 2024 paper).

6. Action Items

#	Action Item	Responsible	Due Date
1	Correct the reference list and in-text citations in the proposal.	Raza	3 Nov 2026
2	Write a one-page summary of each of the six key papers.	Raza	10 Nov 2026
3	Keep the metric definitions in the docs folder for the final report.	Raza	30 Oct 2026

7. Other Notes

- Sir recommended keeping a small reading log (paper, main idea, how it helps my project).

8. Next Meeting

Tuesday, 3 November 2026 - Month 1 review, risks and plan for FN-03.